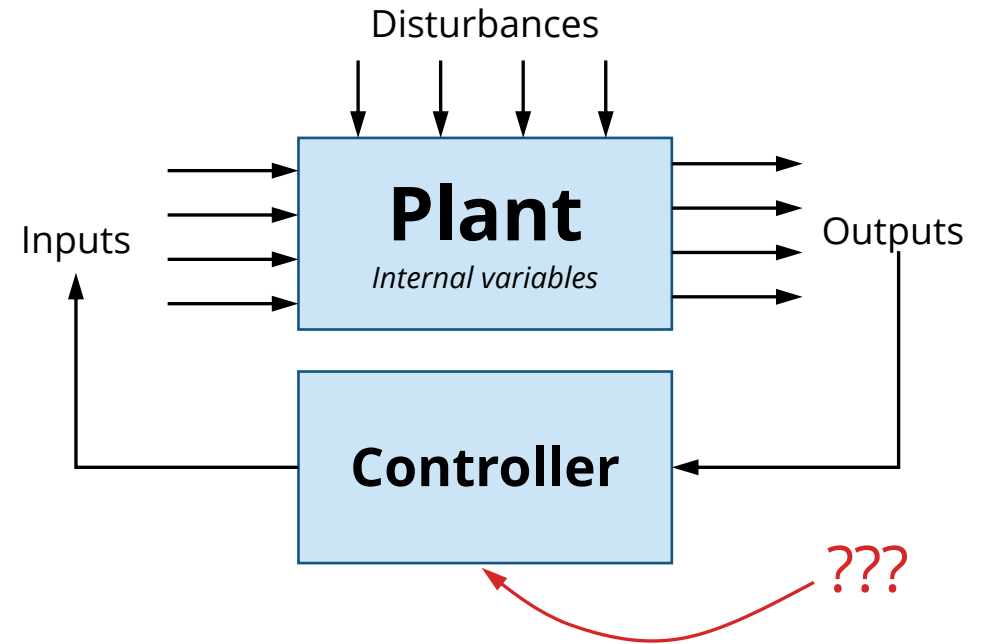




AA 513 Multivariable Control

Lecture 03 // Week 3

Instructor: Prof. Karen Leung

Announcements

- Exercises due Thursday
- TA OH Wed 6PM
- Extra credit for Ed Discussion participation
 - Great discussion so far!
- Released Matlab – Python “Rosetta Stone”
- Ed Discussion
 - Share error message, and python “code snippet”

Last time

- Optimization
 - Gradient descent
 - Convex optimization
 - Frame control problems as (convex) optimization problems
 - Introduced CVXPY

This week

- Lyapunov theory
- Invariant sets
- Control barrier functions
- <https://uw-ctrl.github.io/lmc-book/lectures/control-certificates.html>

Exercises

- Ex4 Equilibria
- Ex5 Lyapunov function
- Ex6 Control Barrier Functions and safety filters

Equilibria

- Given dynamics $\dot{x} = f(x)$, we want to find *equilibrium points*
 - x^{eq} such that $\dot{x} = 0$
- Often define operating points for control design
- Reveal qualitative behavior of nonlinear systems
 - Does the system converge to or diverge from equilibrium points?
- We find equilibrium points because they anchor our understanding of what the system does in steady state and form the basis for analyzing and controlling its behavior.

Stability

- Stability is an important property to characterize.
- Can tell us whether the system's behavior is predictable, safe, and controllable in the face of small perturbations or disturbances.
- If “do nothing”, will the system “stay” or “explode/fall”?
 - E.g., Chemical processes, position of aircraft

Types of stability: Regarding a system I/O

BIBO stability: A system is BIBO stable if every bounded input produces a bounded output.

- *If you feed it a finite signal, it produces a finite response — no unbounded growth or oscillation.*

- **BIBO stability check:** For a rational transfer function (LTI systems): *ALL* poles have negative real parts

- $G(s) = \frac{1}{s^2 + \omega^2}$ BIBO stable?

Types of stability: Regarding equilibria

- **Lyapunov stable:** If trajectory starts close to equilibrium point, then trajectories stay close
 - No guarantees on convergence
- **Asymptotic stability:** If trajectory starts close to equilibrium point, then trajectories eventually converge to the equilibrium point
- **Exponential stability:** If trajectory starts close to equilibrium point, then trajectories converge to the equilibrium point at an *exponential rate*.

Types of stability: Local vs Global

- **Local stability:** Stability property only holds *near* equilibrium point.
 - Common for nonlinear systems
- **Global stability:** Stability property holds *for all* initial conditions
 - Often harder to prove
 - Easy to prove for linear systems

Stability of linear systems

- Consider $\dot{x} = Ax$, where solution is: $x(t) = e^{At}x(0)$
- Stability analysis based on properties of A
- $A = PDP^{-1} \Rightarrow e^A = Pe^{Dt}P^{-1} = P \begin{bmatrix} e^{\lambda_1 t} & 0 \\ 0 & e^{\lambda_n t} \end{bmatrix} P^{-1}$
- Asymptotic stability depends on eigenvalues. All real parts negative implies all trajectories will converge to 0 for any initial condition.
- **Discrete-time LTI:** Eigenvalues lie inside of the unit circle

What about nonlinear systems?

- What about $\dot{x} = f(x)$?
- Linear algebra can't save us ☹️
- All nonlinearities are unique

Lyapunov analysis (Rough idea)

- Consider an energy-like function associated with the state trajectory centered about equilibrium
- Energy = 0 implies at equilibrium point
- If the energy is **always decreasing** along system trajectories, then the trajectories must approach the equilibrium
- If the energy is **non-increasing** along system trajectories, then the trajectories do not diverge from equilibrium
- Works for nonlinear systems!

Lyapunov theorem

i Theorem (Lyapunov function)

Given a general nonlinear dynamical system $\dot{x} = f(x)$, $x \in \mathcal{X} \subset \mathbb{R}^n$. Without loss of generality, let $x = 0$ be an equilibrium state. Let $V : \mathbb{R}^n \rightarrow \mathbb{R}$ be a scalar function that is continuous, and has continuous first derivatives. Then the origin $x = 0$ is Lyapunov stable if:

- $V(0) = 0$
- $V(x) > 0 \forall x \in \mathcal{X} \setminus \{0\}$
- $\dot{V}(x) = \nabla V(x)^T f(x) \leq 0 \forall x \in \mathcal{X}$

How does last equation change for Lyapunov/asymptotic/exponential stability?

Finding a Lyapunov function is not always easy.

Lyapunov stability theorem of linear systems

- For a linear system $\dot{x} = Ax$, consider a *quadratic* Lyapunov candidate

$$V(x) = x^T P x, \quad P \succ 0$$

P positive definite and symmetric

- Compute $\dot{V}(x(t))$
$$= \dot{x}^T P x + x^T P \dot{x}$$
$$= x^T A^T P x + x^T P A x$$

Choose a Q, and solve for P

$$x^T (A^T P + P A) x < 0 \Rightarrow A^T P + P A + Q = 0, Q \succ 0$$

Discrete time Lyapunov equation

[exercise] What is the discrete-time Lyapunov equation?

$$V(x_t) = x_t^T P x_t, \quad \begin{matrix} x_{t+1} = Ax_t \\ V(x_{t+1}) - V(x_t) < 0 \end{matrix}$$

What if there is a control input?

- What happens when $\dot{x} = f(x, u)$? The control input could make the system stable/unstable.
- **Pole placement.** For linear systems, $\dot{x} = Ax + Bu$, consider full state feedback linear controller $u = -Kx$
 - Closed loop system: $\dot{x} = (A - BK)x$
 - Choose K that makes the closed-loop eigenvalues have negative real-parts
 - (See exercise)

What if there is a control input?

- **Control Lyapunov Function**

- Given x , choose $u \in U$ such that

$$\nabla V(x)^T f(x, u) \leq -\alpha(V(x)) \quad (\text{or } 0, \text{ or } <)$$

- What happens if dynamics are control affine?
 - Ensure feasible control for all states. May need to adjust α

- **Intuition:** Pick control satisfying the constraint $\dot{V}(x) \leq -\alpha(V(x))$

Control Lyapunov Function example

- $\dot{x}_1 = x_2, \dot{x}_2 = -x_1 + u, V(x) = \frac{1}{2}(x_1^2 + x_2^2),$

Exercises

Then control barrier functions + exercises

Control invariant sets

- **Forward invariant set:** A set such that if system starts inside, it will remain inside.
 - Pendulum
- **Control invariant set:** Same as above, but “there exists a control” such that...
- Instead of converging to state/point, can we remain inside a set?

Level sets of (control)Lyapunov functions are (control) invariant sets

Control Barrier Functions

i Definition (Control barrier function)

Given a set $\mathcal{S} = \{x \mid b(x) \geq 0\}$ where $b : \mathbb{R}^n \rightarrow \mathbb{R}$. Then b is a control barrier function for a system $\dot{x} = f(x, u)$ if there exists a class $\mathcal{K}$ function α such that

$$\max_{u \in \mathcal{U}} \nabla b(x)^T f(x, u) \geq -\alpha(b(x)), \quad \forall x \in \mathcal{S} \quad (3)$$

- Barrier function describes set to keep system inside of.
- Constraint that *bounds the rate* at which the system approaches the boundary.
- $\dot{b}(x) \geq 0$ when $b(x) = 0$
- What happens when dynamics are control affine?

CBF safety filter

- We can use CBFs to *filter*, or adjust, the desired control to ensure safety (and stability) conditions are satisfied.
- For current state x , and desired control input $u^{desired}$ (e.g., from high-level planner)

$$\begin{aligned} & \min_u ||u - u^{desired}||_2^2 \\ \text{s.t. } & \nabla b(x)^T (f_0(x) + B(x)u) \geq -\alpha(b(x)) \\ & |u| \leq u^{limit} \end{aligned}$$

Question: What type of optimization problem this is?

CBF-CLF filter

- We can add a CLF constraint as well!

$$\begin{aligned} & \min_u \|u - u^{desired}\|_2^2 \\ \text{s.t.} \quad & \nabla b(x)^T (f_0(x) + B(x)u) \geq -\alpha_{CBF}(b(x)) \\ & \nabla V(x)^T (f_0(x) + B(x)u) \leq -\alpha_{CLF}(b(x)) \\ & |u| \leq u^{limit} \end{aligned}$$

Problems?

Further readings

- See course notes:
 - <https://uw-ctrl.github.io/lmc-book/lectures/control-certificates.html#other-topics-and-extra-readings>
- Use neural networks to parameterize functions
- Learn CBFs from demonstrations
- Used to filter neural network policies
- Used as inductive bias in learning-based models
- Professor Aaron Ames from Caltech:
https://www.youtube.com/watch?v=ZC3T_P_8xpE